Physics of Complex Systems Homework 9

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Problem 1 (The First Gap)**.** The Hamiltonian of the Ising quantum chain is defined by

$$H_N = \sum_{n=0}^{N-1} \left(\sigma_n^x \sigma_{n+1}^x + \lambda \sigma_n^z \right),$$

where $\lambda > 0$ and

$$\sigma_n^{x|y|z} = 1_{2^n} \otimes \sigma^{x|y|z} \otimes 1_{2^{N-n-1} \times 2^{N-n-1}}$$

are Pauli matrices acting at site n, assuming periodic boundary conditions. The first (or lowest) gap Δ_N is defined as the difference between the first two lowest-lying eigenvalues of H_N.

(a) In Markov processes, the first gap of $\mathcal{L}$ is the inverse leading relaxation time. In particle physics, the first gap measures the mass. What is the interpretation of the first gap of H_N in the present context?

(b) With the *Mathematica*® code fragment for the tensor product $\otimes$ given in the lecture notes,[1] write a code snippet to set up the Hamiltonian for arbitrary N and λ.

Cross-check: The eigenvalues for $N = 3$ (i.e. 8×8) are $\{-\lambda - 1,\ -\lambda - 1,\ \lambda - 1,\ \lambda - 1,\ -2\sqrt{\lambda^2 - \lambda + 1} + 1,\ 2\sqrt{\lambda^2 - \lambda + 1} + 1,\ -2\sqrt{\lambda^2 + \lambda + 1} + 1,\ 2\sqrt{\lambda^2 + \lambda + 1} + 1\}$.

(c) With (b), compute numerically the first gap of H_N for

$$N = 2, 4, 6, 8, 10, 12$$

and plot the gap for $\lambda = 0.8,\ 1,\ 1.2$ (double-logarithmically) as a function of N. For which λ do you get an almost straight line of points and why?[2]

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[1] You are of course free to use any other algebraic computer system as well.

[2] The execution for $N = 12$ should take less than 20 seconds. If not, go only up to $N = 10$.

(d) Compute the negative discrete logarithmic derivative

$$\nu(N) = -\frac{\ln[\Delta_{N+2}/\Delta_N]}{\ln[(N+2)/N]}$$

for $\lambda = 1$ and

$$N = 2, 4, 6, 8, 10,$$

and guess the value of the critical exponent ν, defined by

$$\nu = \lim_{N\to\infty} \nu(N).$$

Proof. (a) The first gap represents the excitation energy between the ground state and the first excited state.

(b) First we define the tensor product on Pauli matrices, using a **one-indexed convention** instead.

```
In[1]:= σ[k_, x_, n_] := KroneckerProduct[IdentityMatrix[2^(k - 1)], PauliMatrix[x],
IdentityMatrix[2^(n - k)]]
```

Then, we define the Ising Hamiltonian, being careful to use a mod function to impose periodic boundary conditions

```
In[2]:= IsingHamiltonian[n_] := Sum[σ[k, 1, n] . σ[Mod[k + 1, n, 1], 1, n] + λ σ[k, 3, n], {k, 1,
 n}]
```

(c) We define a function to find a spectral gap, given a list of eigenvalues

```
In[3]:= spectralgap[lst_] := (Abs[#[[1]] - #[[2]]] &)[TakeSmallest[lst, 2]]
```

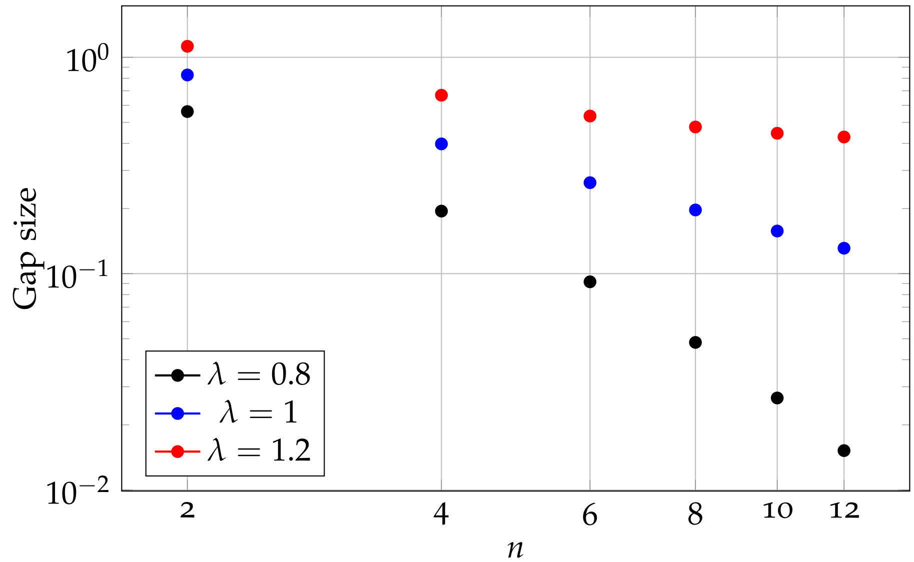
and finally we compute the spectral gaps as a function of n for the various λ.

```
In[4]:= λ12 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 1.2], {n,
2, 12,2}], n];

In[5]:= λ1 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 1.], {n, 2,
 12,2}], n];

In[6]:=  λ08 = spectralgap /@ Monitor[Table[Eigenvalues[IsingHamiltonian[n] /. λ → 0.8], {n
, 2, 12,2}], n];
```

The results are as follows



The line $\lambda = 1$ has an almost linear trend. In a log log plot, the trend appears linear $\log y = a \log x + b$ when the dependence is polynomial, i.e. $y = Ax^B$.

Since $\lambda = 1$ is the critical point, order parameters such as the gap size take power law scaling relations, yielding the approximately linear trend in the log log plot.

(d) We compute the discrete derivative using the following Mathematica code

```
In[7]:=  Table[-Log[λ1[[i + 1]]/λ1[[i]]]/Log[nlst[[i + 1]]/nlst[[i]]], i, 1, Length[λ1 - 1]]
```

In the convention given in the problem, this yields values

N	2	4	6	8	10
$\nu(N)$	1.05824	1.01784	1.00874	1.0052	1.00345

The guess would be that $\nu(N) \xrightarrow{N\to\infty} 1$. In particular, this implies that

$$\Delta_N \to \frac{1}{N}$$

which can be easily checked by substituting this expression back into the definition of the discrete logarithmic derivative. $\square$

Problem 2 (DP Mean Field Approximation)**.** In the mean field approximation of directed percolation (DP) extended by a term for diffusion, the density of active sites $\rho(x,t)$ evolves according to the partial differential equation

$$\dot{\rho} = \lambda\rho(1-\rho) - \rho + D\nabla^2\rho.$$

(a) Determine the homogeneous (space-independent) stationary (time-independent) density ρ_{stat} as a function of λ. Where is the critical point λ_c? Test the solutions for stability by a stability analysis.

(b) Find the exponent for

$$\rho_{\text{stat}} \sim (\lambda - \lambda_c)^\beta$$

near the critical point.

(c) At criticality $\lambda = \lambda_c$, the homogeneous density $\rho(t)$ decays as

$$\rho(t) \sim t^{-\delta}.$$

Compute the exponent δ.

(d) Show that the Green's function of the mean field equation at the critical point can be written in the form

$$G(\mathbf{x}, t) = t^{-1} f\left(\frac{x^2}{t}\right).$$

Hint: It is sufficient to derive an autonomous differential equation for f; you do not have to compute f explicitly. Restrict yourself to $t > 0$ and ignore the scaling of the δ-functions.

Proof. (a) If the density is homogeneous, the left hand side, as well as the second term on the right hand side vanishes and we are left with

$$\lambda\rho_{\text{stat}}(1-\rho_{\text{stat}}) - \rho_{\text{stat}} = 0.$$

We factor ρ out to find

$$0 = \rho_{\text{stat}}[\lambda(1-\rho_{\text{stat}}) - 1]$$

which has either the solution $\rho_{\text{stat}} = 0$ or $\lambda(1-\rho_{\text{stat}}) - 1 = 0$, which implies

$$\rho = 1 - \frac{1}{\lambda}.$$

Note that on physical grounds the static density always has to be positive; hence, the second is only a valid solution for $\lambda > 1$.

We perform a stability analysis on $\rho_{\text{stat}} = 0$ by setting $\rho(x,t) = \delta\rho(x,t)$ with $\delta\rho(x,t)$ small. Then, the equation of motion simplifies to

$$\delta\dot{\rho}(x,t) = (\lambda - 1)\delta\rho + D\nabla^2\delta\rho(x,t)$$

We will ignore the Laplacian term for now; a proper justification will come later. We then have

$$\delta\dot{\rho} = (\lambda - 1)\delta\rho$$

This is a harmonic oscillator equation if $\lambda < 1$, meaning that this solution is stable in that case. However, if $\lambda > 1$, the perturbation does not vanish, but instead grows exponentially. Thus, this solution is unstable for $\lambda > 1$.

For the solution $\rho_{\text{stat}} = 1 - \frac{1}{\lambda}$, we again expand

$$\begin{aligned}
\delta\dot{\rho} &= \lambda\left(\rho_{\text{stat}} + \delta\rho\right)\left(1 - (\rho_{\text{stat}} + \delta\rho)\right) - (\rho_{\text{stat}} + \delta\rho) \\
&= -\delta\rho + \delta\rho\lambda - \rho_{\text{stat}} + \lambda\rho_{\text{stat}} - 2\rho_{\text{stat}}\lambda\delta\rho - \lambda\rho_{\text{stat}}^2 + O(\delta\rho^2) \\
&= \lambda\rho_{\text{stat}}(1 - \rho_{\text{stat}}) - \rho_{\text{stat}} + \delta\rho(-1 + \lambda - 2\rho_{\text{stat}}\lambda) \\
&= \delta\rho(-1 + \lambda - 2\rho_{\text{stat}}\lambda)
\end{aligned}$$

where we have used the fact that ρ_{stat} is an equilibrium density, and thus $\lambda\rho_{\text{stat}}(1 - \rho_{\text{stat}}) - \rho_{\text{stat}}$ vanishes. Then, substituting in explicitly $\rho_{\text{stat}} = 1 - \frac{1}{\lambda}$, we find

$$\delta\dot{\rho} = (1 - \lambda)\delta\rho$$

Again, this is a harmonic oscillator, but for $\lambda > 1$ instead, showing that the solution $\rho_{\text{stat}} = 1 - \frac{1}{\lambda}$ is stable along all of $\lambda > 1$, where it exists.

(b) Since $\lambda_c = 1$, we define $\delta\lambda = \lambda - \lambda_c$ and expand in powers of $\delta\lambda$:

$$\begin{aligned}
\rho_{\text{stat}} &= 1 - \frac{1}{\lambda} \\
&= 1 - \frac{1}{\lambda_c + \delta\lambda} \\
&= 1 - \frac{1}{\lambda_c} + \frac{\delta\lambda}{\lambda_c^2} + O(\delta\lambda^2)
\end{aligned}$$

$$= \frac{\delta\lambda}{\lambda_c^2} \qquad\qquad \lambda_c = 1$$
$$= \frac{1}{\lambda_c^2}(\lambda - \lambda_c)^1$$

Thus the critical exponent $\beta = 1$.

(c) We directly substitute in $\lambda_c = 1$ to arrive at the mean field equation

$$\dot{\rho} = -\rho^2 + D\nabla^2\rho$$

Since the density is homogeneous, the Laplacian vanishes and wee are left with

$$\dot{\rho} = -\rho^2$$

This is a simple ordinary differential equation that can be solved directly using separation of variables.

$$\begin{aligned}
\dot{\rho} &= -\rho^2 \\
\frac{\mathrm{d}\rho}{\rho^2} &= -\,\mathrm{d}t \\
-\frac{1}{\rho} &= -t - A \\
\rho &= \frac{1}{t + A}
\end{aligned}$$

However, we know that the scaling law is only expected to be accurate at large times. Hence, we have

$$\rho(t) \xrightarrow{t\to\infty} \frac{1}{t}.$$

By comparison, this yields $\delta = 1$.

(d) We first linearise the equation; now, the task is to find the Green's function of the linear ODE

$$(\partial_t - D\nabla^2)\rho = 0$$

This must be done since nonlinear differential equations do not admit Green's functions at all. The Green's function is defined through the expression

$$LG(\vec{\mathbf{x}}, t) = \delta(\vec{\mathbf{x}})\delta(t)$$

where L is a differential operator, in this case $\partial_t - D\nabla^2$.

Since we are only working for $t > 0$, the delta function vanishes, and we are actually just looking for solutions of

$$LG(\vec{\mathbf{x}}, t) = 0$$

We will directly substitute the ansatz in the problem to show that the ansatz yields a good Green's function. First, we compute some derivatives

$$\begin{aligned}
\partial_t \left[\frac{1}{t} f\left(\frac{x^2}{t}\right)\right] &= -\frac{1}{t^2} f\left(\frac{x^2}{t}\right) + \frac{1}{t} f'\left(\frac{x^2}{t}\right)\left(-\frac{x^2}{t^2}\right) \\
\partial_x \left[\frac{1}{t} f\left(\frac{x^2}{t}\right)\right] &= \frac{1}{t} f'\left(\frac{x^2}{t}\right)\frac{2x}{t} \\
\partial_x^2 \left[\frac{1}{t} f\left(\frac{x^2}{t}\right)\right] &= \frac{2}{t^2} f'\left(\frac{x^2}{t}\right) + \frac{2x}{t^2} f''\left(\frac{x^2}{t}\right)\frac{2x}{t}
\end{aligned}$$

Substituting all these derivatives in, we find that f satisfies the equation

$$-\frac{1}{t}^2 f\left(\frac{x^2}{t}\right) - \frac{x^2}{t^3} f'\left(\frac{x^2}{t}\right) = D\left[\frac{2}{t^2} f'\left(\frac{x^2}{t}\right) + \frac{4x^2}{t^3} f''\left(\frac{x^2}{t}\right)\right].$$

Thus any function of the form given in the problem with f satisfying the differential equation are a valid Green's function. To prove the converse, we note that the heat equation has rotational invariance and scaling invariance if we scale $t \mapsto b^2 t$ and $x \mapsto bx$, with the difference in powers arising from the Laplacian being 2nd order in the derivative. $\square$